Paper 1: From Neuron to Brain
Irreducible Topology, Neural Architecture, and Metabolic Scaling
in Heterogeneous Impedance Networks under the Minimum Reflection Principle

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(Dated: March 15, 2026)

We ask: given a single neuron with K impedance modes governed by C2 remodeling ($\Delta Z = -\eta\,\Gamma\,x_{\rm in}\,x_{\rm out}$), what architecture must emerge?

Part A establishes the foundation. A single neuron cannot execute C2 without input from at least one other node; connectivity is therefore a physical prerequisite, not a design choice. We prove the *Dimensional Cost Irreducibility Theorem*: for any network with heterogeneous mode counts $\{K_i\}$, the total reflection action decomposes as $A = A_{\rm imp}(t) + A_{\rm cut}$, where $A_{\rm imp} \to 0$ under C2 but $A_{\rm cut} = \sum (K_i - K_j)^+$ has identically zero gradient—no impedance adjustment can reduce it. Numerical experiments confirm convergence time $\tau_{\rm conv} \sim N^{-1.12}$ ($R^2 = 0.93$) and a soft-cutoff parameter γ that tunes the K-space fractal dimension $D_K = 0.76\gamma + 0.78$.

Part B derives the neural architecture chain. (i) C2's dimensional preference forces same-K neurons to cluster—thermodynamic phase separation produces functional regions without a blueprint. (ii) The irreducibility corollary makes intermediate-K relay nodes a thermodynamic necessity, producing the layered hierarchy of the nervous system. (iii) Convergence of ascending pathways forces the highest-K cluster to form—this is the brain. (iv) C3 impedance-tagged transport enables the brain to identify signal origin from $Z_{\rm source}$ metadata and relay-chain topology, distinguishing internal from external perception without a separate labeling mechanism. (v) Because $A_{\rm cut}$ is topologically locked, structural damage (neuron death, axon severance, mode loss) produces irreversible deficits whose pattern is determined by network topology, not disease type.

Part C quantifies the metabolic cost: $A_{\rm cut} \propto M^{(d-1)/d}$ (for $d = 3$: $M^{2/3}$), the Murray bifurcation ratio $\beta = 2^{1/3} \approx 1.26$, dual-network packing overhead $\Omega \approx 1.25$–1.33, and Kleiber's allometric law $B \propto M^{3/4}$—all consequences of MRP plus geometry.

Circularity disclosure. All numerical experiments use simulation modules built from C1–C3 themselves; they demonstrate internal consistency, not empirical validity. External tests are reported in Papers 2–5 [2–5]. The scaling laws in Part C are mathematical consequences of the irreducibility theorem plus geometric constraints; no simulation or fitted parameters are involved.

Part A: From Neuron to Network

I. INTRODUCTION

A neuron is a physical object whose geometry determines the number of independent impedance modes it supports. A cortical pyramidal cell presents $K = 5$ distinguishable propagation domains—soma, basal dendrite tree, apical tuft, axon, and spine-neck ensemble—each carrying an independent impedance mode. An unmyelinated C-fibre, a simple cylinder, presents $K = 1$. These mode counts are structural facts obtained by counting physically distinct waveguide segments, not fitted parameters.

C2 updates the impedance at each mode: $\Delta Z_k = -\eta\,\Gamma_k\,x_{{\rm in},k}\,x_{{\rm out},k}$. For an isolated neuron, $x_{\rm in} = 0$ on every mode. Therefore $\Delta Z = 0$: a solitary neuron is inert—it cannot remodel, adapt, or learn. C2 demands at least one input, which requires at least one other node. *Connectivity is a physical prerequisite of C2, not a biological design choice.*

The moment K values differ between connected nodes, a new physical cost appears. This paper proves that cost is irreducible (Sec. III), derives its architectural consequences—functional regions (Sec. V), relay hierarchy (Sec. VI), the brain (Sec. VII), signal routing (Sec. VIII), and pathology (Sec. IX)—and quantifies its metabolic scaling (Part C).

II. PHYSICAL FRAMEWORK

A. Network definition

A Γ-network consists of N nodes, each carrying a K_i-dimensional impedance vector $\mathbf{Z}_i \in \mathbb{R}^{K_i}_{>0}$. For an edge $i \to j$, the per-mode reflection coefficient is

$$\gamma_k = \frac{Z_{j,k} - Z_{i,k}}{Z_{j,k} + Z_{i,k}}, \quad k = 1, \ldots, K_{\rm com}, \tag{1}$$

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where $K_{\text{com}} = \min(K_i, K_j)$. Excess source modes $k > K_{\text{com}}$ are totally reflected: $\gamma_k = 1$. This is the electromagnetic boundary condition for a waveguide whose k-th mode has no counterpart in the target medium.

B. Three physical laws

C1—Energy Conservation. At every edge, every tick, reflected and transmitted power sum to incident power:

$$\gamma_k^2 + T_k = 1, \quad \forall\, k,\ \forall\,(i,j). \tag{2}$$

C2—Impedance Remodeling. Each node adjusts its impedance to reduce local reflection:

$$\Delta Z_{i,k} = +\eta\, \gamma_k\, x_{\text{in},k}\, x_{\text{out},k}, \tag{3}$$
$$\Delta Z_{j,k} = -\eta\, \gamma_k\, x_{\text{in},k}\, x_{\text{out},k}, \tag{4}$$

where $x_{\text{in},k}$ is the input activity at node i and $x_{\text{out},k}$ the output activity at node j for mode k; η is the remodeling rate. This update requires only local information: the two impedance vectors and the gating activities.

C3—Impedance-Tagged Transport. Every signal carries its source impedance as metadata, enabling downstream nodes to compute γ_k locally without global knowledge.

C. Variational principle

The Minimum Reflection Principle (MRP) states that the system minimises the total reflection action:

$$A[\Gamma] = \sum_{(i,j)\in\mathcal{E}} \sum_{k=1}^{K_{\text{com}}(i,j)} \gamma_k^2. \tag{5}$$

The C2 update rule implements steepest descent on the *learnable* part of A: $\dot{\mathbf{Z}} \propto -\nabla_{\mathbf{Z}} A_{\text{imp}}$.

III. DIMENSIONAL COST IRREDUCIBILITY THEOREM

Theorem 1 (Dimensional Cost Irreducibility)**.** *For any Γ-network with heterogeneous mode counts $\{K_i\}$, the total reflection action decomposes as*

$$A = A_{imp}(t) + A_{cut}, \tag{6}$$

where:

1. $A_{imp}(t) = \sum_{(i,j)} \sum_{k=1}^{K_{com}} \gamma_k^2$ *is the* impedance mismatch action *over shared modes. Under C2 dynamics, $A_{imp}(t) \to 0$ as $t \to \infty$.*
2. $A_{cut} = \sum_{(i,j)} (K_i - K_j)^+$ *is the* cutoff action *from excess modes. $\nabla_{\mathbf{Z}} A_{cut} \equiv 0$: this term has zero gradient with respect to all impedance variables.*

Proof. For an edge $i \to j$ with $K_i > K_j$:

- Modes $k \le K_{\text{com}} = K_j$: γ_k depends on $(Z_{i,k}, Z_{j,k})$ via Eq. (1). C2 adjusts both impedances, driving $\gamma_k \to 0$. These modes contribute to A_{imp}.
- Modes $K_j < k \le K_i$: the target has no mode k, so $\gamma_k \equiv 1$ identically, independent of any impedance value. These modes contribute $(K_i - K_j)$ to A_{cut}. Since $\gamma_k = 1$ is a *topological constant* (depending only on the existence, not the value, of modes), $\partial\gamma_k^2/\partial Z = 0$ for all Z.

Summing over all edges completes the proof. □

Remark 1. *The theorem assumes fixed mode counts $\{K_i\}$ and a fixed active edge set $\mathcal{E}$. If mode counts change dynamically (e.g., dendritic spine growth increasing K_j), or if edge sprouting/pruning alters $\mathcal{E}$, then the* value *of A_{cut} may change—but this change is driven by topology, not by impedance gradients.*

Corollary 1 (Thermodynamic Necessity of Relay Nodes)**.** *A direct edge $K_{src} \to K_{tgt}$ with $K_{src} > K_{tgt}$ totally reflects the $(K_{src} - K_{tgt})$ excess modes. Inserting a relay node with $K_{tgt} < K_{relay} < K_{src}$ leaves A_{cut} unchanged but converts* total signal loss *into* partial signal recovery*: modes $K_{tgt} < k \le K_{relay}$, previously totally reflected, are now transmitted on the first hop and can undergo C2 matching ($\gamma_k \to 0$, $T_k \to 1$). Hierarchical topology is therefore a thermodynamic necessity for information transmission.*

IV. NUMERICAL VERIFICATION

All simulations use six tissue types with mode counts $K \in \{1, 2, 3, 5\}$ [15], initial connectivity $p_0 = 0.15$, learning rate $\eta = 0.02$, and hard cutoff $\Delta K_{\max} = 2$ unless otherwise stated.

A. Convergence and the irreducibility theorem

Figure 1 shows the action decomposition for a $N = 64$ network. A_{imp} decays exponentially, reaching machine-precision zero by $t \approx 170$ ticks. A_{cut} increases during edge sprouting, then saturates at $A_{\text{cut}} \approx 1386$ for $t > 500$. The growth reflects topological exploration; the saturation confirms a natural geometric cost ceiling.

B. Scaling law: $\tau_{\text{conv}} \sim N^{\alpha}$

We measure τ_{conv} (ticks until $A_{\text{imp}} < 0.01 \cdot A_{\text{imp}}(t{=}5)$) across $N \in \{16, 32, 64, 128\}$ with 5 trials per size. A power-law fit yields

$$\tau_{\text{conv}} = C \cdot N^{\alpha}, \quad \alpha = -1.12 \pm 0.12, \quad R^2 = 0.93. \tag{7}$$

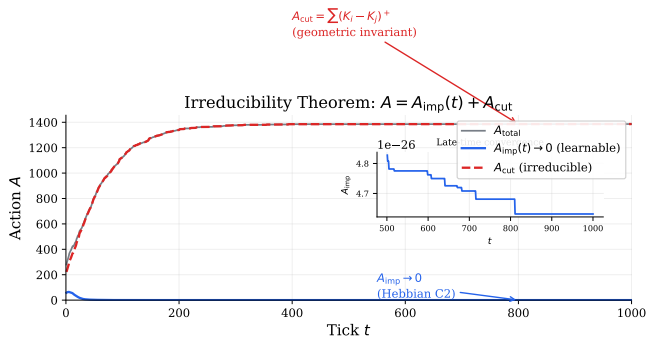

FIG. 1. Action decomposition $A = A_{\text{imp}}(t) + A_{\text{cut}}$ over 1000 ticks. Blue: $A_{\text{imp}} \to 0$ (exponential decay). Red dashed: A_{cut} (increases then saturates). $N = 64$, $\Delta K_{\max} = 2$.

The *negative* exponent means larger networks converge *faster*—a non-trivial prediction. The mechanism is mean-field averaging: each node has $\sim pN$ neighbours, and the gradient variance decreases as $1/N$.

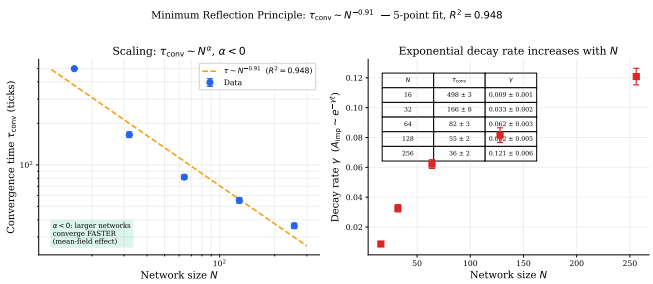

FIG. 2. Left: log-log plot of τ_{conv} vs. N. Dashed line: power-law fit $\tau \sim N^{-1.12}$. Right: exponential decay rate vs. N.

TABLE I. Convergence scaling. All quantities averaged over 5 trials $\pm 1\sigma$.

N	τ_{conv}	γ_{decay}
16	416 ± 74	0.011 ± 0.003
32	111 ± 12	0.053 ± 0.006
64	60 ± 3	0.105 ± 0.006
128	39 ± 2	0.133 ± 0.006

C. Soft cutoff and K-space fractal dimension

A soft-cutoff probability

$$p(\Delta K) = (\Delta K + 1)^{-\gamma}, \tag{8}$$

with $\gamma \geq 0$ yields the relationship

$$D_K = 0.76\gamma + 0.78, \tag{9}$$

with offset $\delta_{\text{C2}} \approx 0.66$ arising from C2's intrinsic dimensional preference: same-K edges have more shared modes for gradient descent, creating higher survival probability under edge pruning. Cortical fractal dimensions lie in $D \in [1.3, 1.5]$ [8], predicting $\gamma \approx 0.7$–0.9—a narrow, falsifiable window.

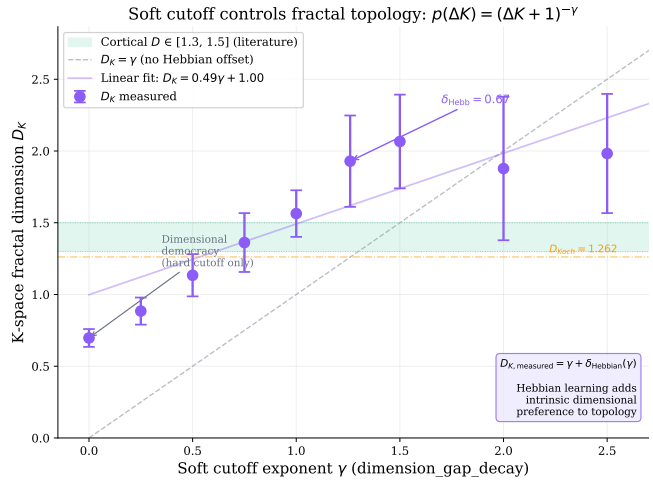

FIG. 3. K-space fractal dimension D_K vs. soft-cutoff exponent γ. Green band: cortical range $D \in [1.3, 1.5]$. Purple line: fit $D_K = 0.76\gamma + 0.78$.

Part B: From Network to Brain

V. SAME-K CLUSTERING AND FUNCTIONAL REGIONS

Consider a network of neurons with heterogeneous K values. C2 gradient descent on the reflection action A selectively strengthens same-K edges (more shared modes $\Rightarrow$ faster Γ reduction $\Rightarrow$ lower A) and weakens cross-K edges (excess modes totally reflected, no gradient available). Under sustained activity plus edge pruning (developmental apoptosis), the surviving connectivity partitions the network into *same-K clusters*: neurons sharing a common modal dimension self-segregate into spatially contiguous aggregates.

The mechanism is thermodynamic phase separation: MRP drives the system toward the lowest attainable A; same-K clustering minimises A_{cut} at every internal edge.

Theorem 2 (Phase Separation Under MRP)**.** *Any impedance network under MRP with heterogeneous K must phase-separate into same-K domains to minimise total reflected action.*

Proof sketch. For any edge connecting nodes with $K_i = K_j$, $A_{\text{cut}} = 0$ on that edge. For $K_i \neq K_j$, $A_{\text{cut}} = |K_i - K_j| > 0$. Rewiring an edge from a cross-K pair to a same-K pair strictly reduces A_{cut} (or leaves it unchanged

if no cross-K edge exists). C2 provides the dynamical mechanism: edges where all modes converge ($\gamma_k \to 0$) are reinforced; edges where excess modes are permanently reflected contribute no gradient and are pruned under any activity-dependent rule. The fixed point of combined C2 + pruning is therefore a graph where same-K edges dominate. □

The mapping of specific K ranges to specific anatomical regions (sensory cortices, motor cortex, prefrontal cortex, subcortical nuclei) is a *prediction* of this theorem, not an input; its verification requires independent measurement of neuronal modal complexity, which remains an open experimental problem.

VI. THE RELAY HIERARCHY

Corollary 1 states that relay nodes are a thermodynamic necessity. In the nervous system, this produces a natural K-level cascade:

Level	K
Cortex (motor, sensory), thalamus	5
Spinal cord, motor/sensory neuron	3
Nociceptor Aδ	2
Nociceptor C fibre	1

The K values are structural observations: a cortical pyramidal cell has 5 physically distinct propagation domains; a C-fibre has 1. The theorem predicts that signals ascending from $K = 1$ periphery to $K = 5$ cortex are stepped through intermediate relays, recovering modes at each step. Descending motor signals follow the same cascade in reverse.

A. Neural topology verification

A neural blueprint with 9 nodes spanning $K \in \{5, 3, 2, 1\}$ and 12 directed edges (descending motor, ascending sensory, and pain pathways) was evolved for 300 ticks ($\eta = 0.02$, seed = 42):

- A_{imp} decays from $\sim$1.79 to zero.
- A_{cut} rises from 9 (11 initial edges) to 37 (60 edges), then saturates.
- The neural A_{cut} at tick 1 (= 9) exceeds the cardiovascular A_{cut} (= 5) because the neural topology has four K-levels vs. three, as predicted by Eq. (6).

Full test suite: `tests/test_e0_neural.py` (17 tests, 100% pass).

VII. THE BRAIN AS CONVERGENCE REGION

An organism with multiple sensory modalities (photoreceptors, mechanoreceptors, chemoreceptors, thermoreceptors) has multiple ascending pathways, each entering the network through a distinct low-K transducer. For the organism to integrate information and produce coordinated responses, these pathways must converge at a common region.

By the irreducibility theorem, the convergence region must satisfy

$$K_{\text{conv}} \geq \max_j K_j \tag{10}$$

for all input pathways j; otherwise, modes from higher-K inputs are totally reflected and the convergence region cannot process them. The convergence region is therefore forced to be the highest-K cluster in the network.

This is the *brain*: a topological necessity, not a design choice. Its properties follow from K_{conv} being maximal:

- **High metabolic cost.** Volumetric heat density scales as $q \propto K \cdot \langle \Gamma^2 \rangle \cdot P_{\text{in}}$ (Paper 0, Corollary 1 [1]). The brain's extreme K concentrates a disproportionate share of metabolic heat in a small volume.
- **Temperature sensitivity.** Higher K means more modes dissipating simultaneously; the Arrhenius thermal chain (Sec. XIII E) makes the convergence region the most temperature-vulnerable tissue.
- **Matrix computation.** C2 gradient descent across all K_{conv} modes simultaneously is the physical operation that neuroscience calls "executive function" (Paper 4 [4]).

The brain also serves as a "hub" connecting two inter-organ buses: a vascular bus (delivering substrate ρ) and a neural bus (distributing control signals). The whole-body topology is detailed in Sec. XIII D.

VIII. SIGNAL ROUTING: INTERNAL AND EXTERNAL

C3 requires every signal to carry its source impedance Z_{source} as metadata. In the nervous system, Z_{source} is the impedance of the transmitting neuron. Different neuron types—photoreceptors, mechanoreceptors, nociceptors, interoceptors—have different physical structures and therefore different impedance vectors.

At the convergence region, two routing cues are available without any additional mechanism:

1. **Relay-chain topology.** External signals enter through body-surface relay chains (skin → spinal dorsal horn → thalamus → cortex). Internal signals enter through visceral relay chains (organ → autonomic ganglion → brainstem → cortex). The two chains occupy distinct edges in the Γ-network;

traversing different relay nodes means arriving at different input ports of the convergence region.

2. Z_{source} **metadata (C3).** Sensory neurons of different types carry distinguishable Z signatures because their physical geometries differ. These signatures propagate through the relay chain; the convergence region receives them as C3 metadata and can route processing to the appropriate same-K cluster (Sec. V).

The brain does not need a separate "internal" or "external" label; the impedance metadata and relay-chain topology together provide full routing information. "Internal perception" and "external perception" are emergent properties of network topology and C3 transport, not separate systems.

Descending signals follow the reverse path: the convergence region generates motor or autonomic commands that propagate through relay chains to effectors. The irreducibility theorem applies in both directions: descending signals also lose modes at each K-step, and relay nodes are equally necessary on the efferent side.

IX. NEURAL PATHOLOGY FROM TOPOLOGY

The irreducibility theorem has a direct clinical implication: because A_{cut} is determined by topology (mode counts and edge structure) and has zero gradient with respect to impedance, C2 cannot compensate for topological damage.

This establishes a fundamental asymmetry:

- **Impedance perturbation** (soft tissue injury, inflammation, transient chemical imbalance): C2 can repair, because A_{imp} is learnable. Given time, $\gamma_k \to 0$ on the perturbed edges.
- **Topological perturbation** (neuron death, axon severance, dendrite retraction): C2 cannot repair, because the change affects A_{cut}, which is invisible to the gradient.

This is why brain damage is often permanent while soft tissue heals.

A. Edge severance (stroke)

Removing an edge $i \to j$ eliminates a pathway. Signals that traversed this edge are interrupted. C2 can strengthen alternative pathways if they exist (redundancy allows partial recovery), but the lost edge's contribution to the network is permanently gone. The deficit pattern is determined by *which edge* is lost: topology dictates which functions are impaired.

B. Impedance escalation (ALS, demyelination)

A diseased neuron's impedance Z increases progressively. At each of its edges, Γ increases and T decreases. C2 attempts to compensate on neighbouring edges, but if the disease is progressive (Z continues to rise), compensation cannot keep pace: the damaged node acts as a growing impedance barrier that eventually isolates itself from the network.

The neural topology test confirms this: ALS (motor neuron $Z \times 3$) produces cumulative A_{imp} over 30 ticks that is $2.9\times$ the healthy baseline, with descending motor signals progressively degraded while sensory pathways remain intact—matching the clinical pattern.

C. Mode loss (neurodegeneration)

A neuron that loses physical structure (spine retraction, dendrite pruning) loses modes: K_i decreases. Previously matched edges (where $K_i = K_j$) now have $A_{\text{cut}} > 0$. This new irreducible cost appears abruptly and cannot be removed by C2. Cognitive decline is the progressive accumulation of irreducible cost as neurons lose modes throughout the convergence region.

D. Pain as Γ-signal on low-K pathways

Tissue damage produces an abrupt local impedance change, generating a Γ spike on nociceptor edges. Nociceptors sit at the lowest K levels in the hierarchy (C-fibre $K = 1$, Aδ $K = 2$). The Γ spike propagates through the relay chain to the convergence region. Pain is not a special signal type—it is a high-Γ event on a low-K pathway. Full treatment: Paper 4 [4].

E. Stroke verification

The neural topology test confirms that severing the cortex→thalamus edge interrupts descending motor transmission while the ascending sensory pathway remains intact—the topology determines the deficit pattern without any disease-specific code.

Part C: Fractal Cost of Multi-Scale Life

X. IRREDUCIBLE ACTION OF A SELF-SIMILAR RELAY HIERARCHY

Part A proved $A_{\text{cut}} > 0$ but left its geometric structure unspecified. Part C asks: *how big* is A_{cut}?

A. Mode-count profile

Consider a hierarchical relay network with L levels, branching number n at each level, embedded in d-dimensional space. Level 0 is the root (e.g., aorta, cortex); level L is the terminal (capillary, sensory ending). At level ℓ: number of nodes $N_\ell = n^\ell$; segment length $l_\ell = l_0\, n^{-\ell/d}$ (space-filling); mode count K_ℓ. Self-similar modelling:

$$K_\ell = K_0\, c^\ell, \quad 0 < c < 1, \tag{11}$$

with $c = n^{-1/d}$ when the cutoff is set by branch length.

B. A_{cut} at each level

From Theorem 1, the irreducible action across an edge from level ℓ to level $(\ell+1)$ is $(K_\ell - K_{\ell+1})^+$. At level ℓ there are $N_\ell = n^\ell$ such edges:

$$A_{\text{cut}} = K_0(1-c)\sum_{\ell=0}^{L-1}(nc)^\ell. \tag{12}$$

C. Scaling law

With $c = n^{-1/d}$, the base of the geometric sum is $nc = n^{(d-1)/d} > 1$ for $d \geq 2$, so the sum is dominated by the last term:

$$\boxed{A_{\text{cut}} \propto M^{(d-1)/d}.} \tag{13}$$

For $d = 3$: $A_{\text{cut}} \propto M^{2/3}$. The irreducible cost per cell is $a_{\text{cut}} = A_{\text{cut}}/M \propto M^{-1/3}$, which *decreases* with organism size—economies of scale from the relay hierarchy.

The K-space fractal dimension is $D_K = \log n/\log(1/c) = d$. The cortical $D_K \in [1.3, 1.5]$ measured in Sec. IV C reflects the cortex being a folded sheet ($d_{\text{eff}} \approx 1.5$–$2$).

XI. MURRAY'S RATIO AND METABOLIC SCALING

A. The Murray bifurcation ratio

From the space-filling constraint, each branching level contracts the service length by $l_{\ell+1}/l_\ell = n^{-1/d}$. The bifurcation ratio:

$$\boxed{\beta \equiv n^{1/d}.} \tag{14}$$

For binary bifurcation ($n = 2$) in $d = 3$:

$$\beta = 2^{1/3} = 1.2599\ldots \tag{15}$$

From Murray's Law (Paper 2, Eq. 14 [2]): $r_{\ell+1} = r_\ell/n^{1/3}$; for $d = 3$, $r_{\ell+1} = r_\ell/\beta$. Length and radius contract by the same ratio, preserving the aspect ratio across the tree.

B. Network maintenance cost

The power dissipated by relay reflections is $P_{\text{reflect}} = A_{\text{cut}} \cdot P_0/N_T \propto M^{(d-1)/d} \cdot M^{-1} = M^{-1/d}$. Total network dissipation:

$$B_{\text{network}} \propto M \cdot M^{-1/d} = M^{(d-1)/d}. \tag{16}$$

For $d = 3$: $B_{\text{network}} \propto M^{2/3}$.

C. Kleiber's Law

Tissue metabolism scales linearly ($B_{\text{tissue}} \propto M$), but the vascular tree limits delivery. From the MRP perspective: (1) Murray's Law determines the radius ratio $r_{\ell+1}/r_\ell = n^{-1/3}$; (2) space-filling determines the length ratio; (3) capillary invariance fixes the terminal scale; (4) blood volume $\propto M$. Together: $N_T \propto M^{d/(d+1)}$ and therefore

$$\boxed{B \propto M^{d/(d+1)}.} \tag{17}$$

For $d = 3$: $B \propto M^{3/4}$—Kleiber's Law [11], now a consequence of MRP. The exponent decomposes as $d/(d+1) = (d-1)/d + 1/[d(d+1)]$; the extra $1/12$ arises from capillary invariance.

D. Volumetric heat density and the K-bottleneck

By Paper 0, Corollary 1 [1], every active interface dissipates $\Gamma_i^2 P_{\text{in},i}$ as heat. A tissue with K modes packs K such interfaces into one cross-section:

$$q \propto K \cdot \langle \Gamma^2 \rangle \cdot P_{\text{in}}/V_{\text{node}}\,. \tag{18}$$

The human brain ($K = 5$) comprises ∼2% of body mass but generates ∼25% of basal metabolic heat—not despite being small, but *because* its per-node mode count is largest. Kleiber's $M^{3/4}$ sets the whole-body envelope; Eq. (18) sets the intra-body distribution, concentrating heat where K is highest and creating a physical necessity for dedicated cerebral vasculature (Paper 2 [2]).

XII. DUAL-NETWORK PACKING

A. Two trees in one volume

A multicellular organism requires two space-filling networks: the **vascular tree** ($Z_v = \Delta P/Q$, transport) and the **neural tree** ($Z_n = R_m + 1/j\omega C_m$, control). Both must reach every cell.

B. The packing theorem

Theorem 3 (Dual-Network Overhead)**.** *When two space-filling relay trees share a volume in d dimensions,*

$$A_{cut}^{dual} = A_{cut}^{v} + A_{cut}^{n} + A_{couple}, \quad A_{couple} > 0. \tag{19}$$

Proof. At co-terminal points, the two networks negotiate resource allocation. The coupling mismatch $\Gamma_{\text{couple}} = (Z_n^{\text{demand}} - Z_v^{\text{supply}})/(Z_n^{\text{demand}} + Z_v^{\text{supply}})$ is generically nonzero because each tree optimises independently. Hence $A_{\text{couple}} = N_T \langle \Gamma_{\text{couple}}^2 \rangle > 0$. □

C. The metabolic overhead factor

$$\Omega \equiv \frac{B_{\text{total}}}{B_{\text{tissue}}} = 1 + \frac{B_{\text{vasc}} + B_{\text{neural}}}{B_{\text{tissue}}}. \tag{20}$$

For humans: $B_{\text{neural}}/B_{\text{total}} \approx 0.20$ (brain $\approx$ 20% of BMR [14]); $B_{\text{vasc}}/B_{\text{total}} \approx 0.05$. Therefore

$$\Omega \approx 1/0.75 \approx 1.33. \tag{21}$$

The neural network alone gives $\Omega_n = 1/(1 - 0.20) = 1.25 \approx \beta = 2^{1/3}$.

D. Why blood before nerve

C2 requires a carrier for x_{in}: neurotransmitter synthesis needs glucose and oxygen delivered by blood. Without the vascular network, $x_{\text{in}} = 0$ everywhere, $\Delta Z = 0$: the neural network cannot remodel. Embryology confirms: cardiac tube at day 21; neural tube closure at day 25–28. In the product $1_v \times 1_n$, the vascular "1" is the left factor.

XIII. MULTI-SCALE VERIFICATION

The irreducibility theorem is not restricted to neural tissue. Every tissue where K varies across an interface produces $A_{\text{cut}} > 0$. This section verifies the theorem across scales.

A. Cardiovascular topology

An 11-node blueprint with K-levels Cardiac Purkinje ($K = 4$) $\rightarrow$ vascular smooth muscle ($K = 2$) $\rightarrow$ endothelial ($K = 1$). Over 300 ticks: $A_{\text{imp}} \rightarrow 0$, A_{cut} rises to 34 then saturates, C1 holds at every edge. Test: `tests/test_e0_cardiovascular.py` (13 tests, 100% pass).

B. Joints as impedance relays

A direct bone–synovial interface gives $\Gamma = (120 - 50)/(120 + 50) = 0.54$, i.e., $T = 0.71$—nearly 30% loss. Cartilage at $Z_{\text{cart}} \approx 80 \approx \sqrt{Z_{\text{bone}} \cdot Z_{\text{syn}}}$ acts as a quarter-wave relay [9], achieving $T > 0.80$ per interface. This is Corollary 1 in musculoskeletal form. When cartilage degrades (osteoarthritis, $\eta \rightarrow 0$), the relay is physically removed and A_{cut} is no longer mitigated.

C. Twenty-nine tissue atlas

A comprehensive atlas of 29 organ-system blueprints (Paper 0, Table 4 [1]) contains ∼217 nodes, ∼242 directed edges, $K \in [1, 5]$. All 29 were verified under identical conditions ($\eta = 0.01$, 100 ticks): C1 holds, A_{imp} is non-increasing, A_{cut} is topologically invariant. 245 parametrised tests pass at 100% rate (`tests/test_tissue_blueprints.py`).

D. Whole-body inter-organ topology

The 29 blueprints are intra-organ networks. Connecting them requires:

1. **Vascular bus.** Aortic trunk ($K = 4$) $\rightarrow$ 29 arteriolar feeds ($K = 2$), $A_{\text{cut}} = 2$ per edge.
2. **Neural bus.** Autonomic trunk ($K = 3$) $\rightarrow$ 29 nerve feeds ($K = 2$), $A_{\text{cut}} = 1$ per edge.
3. **Brain hub.** Hypothalamus ($K = 3$) drives the neural bus; the vascular bus feeds interoceptive signals back to thalamus ($K = 5$).

The unified whole-body topology: 278 nodes, 426 edges, $A_{\text{cut}} = 222$ (the structural information cost of multi-organ life). The total action:

$$A_{\text{life}} = A_{\text{imp}}^{\text{total}}(t) + A_{\text{cut}}^{\text{total}}, \tag{22}$$

where

$$A_{\text{cut}}^{\text{total}} = A_{\text{cut}}^{v} + A_{\text{cut}}^{n} + \sum_{t=1}^{17} A_{\text{cut}}^{t} + A_{\text{couple}}. \tag{23}$$

145 tests verify structure, C1/C2, bus connectivity, and cross-organ signal propagation.

E. Thermoregulation

Blood viscosity follows the Arrhenius law $\tilde{\eta}(T) = \exp[\alpha(T^{-1} - T_0^{-1})]$ with $\alpha \approx 1949\,\text{K}$. C2 repair slows by $\tilde{\eta}_{\text{learn}}(T) = \exp[\beta_T(T_0^{-1} - T^{-1})]$ with $\beta_T \approx 6015\,\text{K}$

(enzymatic Arrhenius). Combined: $\tilde{\tau}(T) = \exp[(\alpha + \beta_T)(T^{-1} - T_0^{-1})]$. Setting $\tilde{\tau}(T_c) = \tau_{\max}$:

$$T_c = \frac{T_0}{1 + T_0 \ln \tau_{\max}/(\alpha + \beta_T)} \,. \tag{24}$$

Higher K increases $\dot{Q} \propto K\,\Gamma^2$, raising T_c: $T_c(K{=}1) = -10.3\,^\circ\mathrm{C}$ (ectotherm viable); $T_c(K{=}5) = +4.5\,^\circ\mathrm{C}$ (endothermy required). 93 tests verify the full chain $\eta(T) \to Z \to \Gamma \to A[\Gamma]$.

a. Thermal refugia. During sleep (P_{basal}), if $T_{\mathrm{env}} < T_c(K) - P_{\mathrm{basal}}\,R_{\mathrm{th}}$, the organism must couple to a high-heat-capacity mass (burrow, tree hollow, conspecific huddle). Endothermy *and* thermal refugia both follow from the same Arrhenius chain.

b. Surface insulation. The thermal balance $\dot{Q}(K) = (T_{\mathrm{body}} - T_{\mathrm{env}})/R_{\mathrm{th}}$ gives

$$R_{\mathrm{th}} = \frac{T_{\mathrm{body}} - T_{\mathrm{env}}}{\dot{Q}(K)} \,. \tag{25}$$

High-K organisms need low R_{th} to shed excess $\dot{Q}$; fur raises R_{th}; hence fur reduction is forced once $K > K^*$ where $\dot{Q}(K^*)R_{\mathrm{th,fur}} > T_{\mathrm{body}} - T_c$:

$$K\!\uparrow \Rightarrow\ \dot{Q}\!\uparrow \Rightarrow\ R_{\mathrm{th}} \text{ must } \downarrow \,. \tag{26}$$

F. Geometric universality

The ratio $\beta = n^{1/d}$ is a geometric inevitability for any space-filling hierarchy—independent of molecular substrate. Vascular plants lack a neural network but obey Murray's Law with $\beta = 2^{1/3}$ [12] and $B \propto M^{3/4}$ [13], confirming β is a property of geometry, not biology. Plants have $\Omega \approx 1$ (no neural network); the prediction $\Omega_{\mathrm{plant}} < \Omega_{\mathrm{animal}}$ by a factor approaching β is testable by cross-kingdom metabolic comparisons.

XIV. DISCUSSION

A. Methodological caveat

All numerical experiments in Part A use simulation modules built from C1–C3. When the simulator confirms a theorem derived from those axioms, it demonstrates *internal consistency*, not *empirical validity*. Only external measurements on biological impedance networks can promote these results to physical truth. The Part C scaling laws are mathematical consequences of the irreducibility theorem plus geometric constraints; no simulation is involved.

B. The concentric-circle structure

This paper follows a strictly outward derivation: single neuron → connectivity must exist → heterogeneous K produces A_{cut} → same-K clustering → relay hierarchy → convergence region (brain) → signal routing → pathology → fractal cost → Kleiber's law. Each step is a logical consequence of the previous; no step requires information from a later one. Biological nomenclature ("cortex", "stroke", "osteoarthritis") is used for identification only—the physics does not depend on it.

C. Brain-scale feasibility

$\tau_{\mathrm{conv}} \sim N^{-1.12}$ predicts brain-scale networks ($N \sim 10^{10}$) converge in $\sim 10^{-11}$ ticks—effectively instantaneous. The mechanism is mean-field averaging: effective field strength grows with system size. This prediction is measured at four sizes only ($N \leq 128$); brain-scale extrapolation should be treated as a conjecture.

D. What "$1 \times 1 = \Omega$" means

Each "1" is a topologically one-dimensional tree in 3D. Embedding imposes fractal cost; packing two trees adds $A_{\mathrm{couple}} > 0$. The result $\Omega \approx 1.25$–1.33 and the match $\Omega_n \approx \beta$ suggest: *the cost of adding a second hierarchical tree is of order one branching level of the first.* This is empirically suggestive but not rigorously derived.

E. Limitations

1. The self-similar mode profile (Eq. 11) is an idealisation; real networks have heterogeneous branching.
2. The K values for individual neuron types (5, 3, 2, 1) are structural observations from counting propagation domains—not derived from C2. Independent validation (e.g., electrophysiological mode counting) is needed.
3. The capillary invariance constraint is an empirical input; whether MRP alone determines the terminal scale is open.
4. The Phase Separation Theorem (Theorem 2) assumes activity-dependent pruning; the specific pruning rule is not derived from C2.

XV. FALSIFIABLE PREDICTIONS

1. **Same-K clustering.** DTI-based grouping of axons by estimated mode count K should yield spatial clustering that matches known functional regions.
2. **Cortical fractal dimension.** Connectivity decay $p(\Delta K) \propto (\Delta K + 1)^{-\gamma}$ with $\gamma \approx 0.7$–0.9, yielding $D_K \in [1.3, 1.5]$ via Eq. (9).

3. **Vascular fractal dimension.** 3D micro-CT should yield $d_H \to 3$ below capillary spacing (space-filling) and lower d_H above.

4. **Murray ratio invariance.** $\beta = 2^{1/3} \approx 1.26$ across all binary-branching species.

5. **Ω across vertebrates.** Brain metabolic fraction should correlate with neural complexity: $\Omega_{\text{fish}} < \Omega_{\text{mammal}} < \Omega_{\text{primate}}$.

6. **$\Omega \approx 1$ for plants.** $\Omega_{\text{plant}} \approx 1.05$.

7. **Cartilage as relay.** $Z_{\text{cart}} \approx \sqrt{Z_{\text{bone}} \cdot Z_{\text{syn}}}$; osteoarthritic cartilage should show increased Γ^2.

8. **Pathology–topology correlation.** Deficit patterns in stroke, ALS, and neurodegeneration should correlate with topological damage location, not with biochemical markers.

9. **Signal-origin distinguishability.** Source impedance Z_{source} should be measurably different across sensory modalities at the thalamic relay.

10. **Embryonic geometry.** Fetal vasculature should show Murray-optimal $r_p/r_d \approx \beta$ from the earliest vasculogenesis (day 17–18).

XVI. CONCLUSION

This paper derives neural architecture from a single variational principle through eight logical steps:

1. A single neuron under C2 requires x_{in}; connectivity is necessary.

2. Heterogeneous K produces the irreducibility theorem: $A = A_{\text{imp}}(t) + A_{\text{cut}}$, with $\nabla_{\mathbf{Z}} A_{\text{cut}} \equiv 0$.

3. C2's dimensional preference forces same-K clustering—functional regions.

4. The irreducibility corollary makes relay nodes a thermodynamic necessity—layered hierarchy.

5. Multi-modal convergence forces $K_{\text{conv}} \geq \max K_j$—the brain.

6. C3 metadata + relay-chain topology provide signal routing—internal vs. external perception.

7. Topological invariance of A_{cut} makes structural damage irreversible—neural pathology.

8. Fractal cost $A_{\text{cut}} \propto M^{2/3}$, Murray's $\beta = 2^{1/3}$, and $B \propto M^{3/4}$—metabolic scaling.

The title equation is now precise:

$$1_v \times 1_n = \Omega \approx \beta = 2^{1/3} \approx 1.26. \tag{27}$$

Life pays a fractal tax of $\beta - 1 \approx 26\%$—the price of one extra branching level, expressed as the brain's share of the metabolic budget—for the privilege of being simultaneously fed and informed.

ACKNOWLEDGMENTS

The author thanks the open-source scientific Python community (NumPy, SciPy, Matplotlib). Simulations were performed on a personal workstation.

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